High-Level Design (HLD) — CredVault

1. Overview

Field	Details
System Name	CredVault
Type	Enterprise Credit Card Management Platform
Architecture Style	Microservices with Event-Driven Communication + Saga Orchestration
Primary Domain	Financial Services / Credit Card Management

Purpose

CredVault is a production-grade credit card management platform that enables users to register, manage credit cards, generate bills, process payments, earn and redeem rewards, and maintain a digital wallet. It serves as a self-learning project demonstrating enterprise microservices patterns: CQRS, Saga orchestration, event-driven communication, API Gateway routing, and the outbox pattern.

Scope

In Scope:

- User registration with email OTP verification and Google OAuth SSO
- Credit card lifecycle management (add, view, manage, soft-delete)
- Automated bill generation and statement creation
- Rewards point system with tier-based earning rates
- Wallet-based and card-direct payments with OTP 2FA
- Distributed transaction management via Saga pattern
- Email notifications for all critical system events
- Admin panel for user, card, billing, and reward management

Out of Scope:

- Real-time payment gateway integration for bill payments (simulated via saga)

- SMS notifications (email only via Gmail SMTP)
- Mobile native applications (web-only via Angular SPA)
- Multi-currency support (USD only)
- Third-party card issuer API integrations

2. Goals & Non-Goals

Functional Goals

1. User registration, email OTP verification, and Google OAuth SSO login
2. Credit card CRUD operations with encrypted storage and soft-delete
3. Bill generation and statement aggregation from card transactions
4. Secure payment processing with OTP 2FA and saga orchestration
5. Rewards system with tier-based earning and redemption during payments
6. Wallet management with Razorpay top-ups
7. Email notifications for registration, OTP, card addition, and payment events
8. Admin capabilities for user, card, billing, and reward tier management

Non-Goals

- Real bank/card network integrations (simulated via internal services)
- Push notifications or mobile app support
- Fraud detection or ML-based risk scoring
- Multi-tenant SaaS architecture (single-tenant design)
- Real-time streaming analytics or dashboards

3. Requirements

Functional Requirements

ID	Requirement	Service
FR-01	User registration with email, name, password	Identity
FR-02	Email OTP verification (6-digit, 10-min expiry)	Identity

ID	Requirement	Service
FR-03	Google OAuth SSO login	Identity
FR-04	Password reset via OTP	Identity
FR-05	User profile management (view, update, change password)	Identity
FR-06	Add/update/delete credit cards (soft-delete)	Card
FR-07	Card number encryption at rest (AES)	Card
FR-08	Record card transactions (purchase, payment, refund)	Card
FR-09	Card issuer management	Card
FR-10	Generate bills for cards	Billing
FR-11	Create statements with transaction breakdown	Billing
FR-12	Reward account management (earn, redeem, reverse)	Billing
FR-13	Reward tier configuration (network/issuer specific)	Billing
FR-14	Initiate payment with OTP 2FA (6-digit, 5-min expiry)	Payment
FR-15	Orchestrate distributed payment via Saga pattern	Payment
FR-16	Wallet management (create, top-up, debit, refund)	Payment
FR-17	Razorpay integration for wallet top-ups	Payment
FR-18	Email notifications for all critical events	Notification
FR-19	Audit logging for all entity changes	Notification
FR-20	Admin user management (list, update status/role, stats)	Identity
FR-21	Admin card management (list, update, view transactions)	Card
FR-22	Admin bill generation and overdue checking	Billing
FR-23	Admin reward tier CRUD	Billing
FR-24	Admin notification and audit log viewing	Notification

Non-Functional Requirements

Requirement	Specification
Scalability	Each service scales independently; stateless services behind API Gateway
Availability	Services are autonomous; one service failure doesn't cascade to others
Consistency	Eventual consistency via RabbitMQ events and Saga compensation
Performance	Async communication for non-critical paths; CQRS for read-heavy operations
Security	JWT Bearer auth, RBAC (User/Admin), AES card encryption, OTP 2FA for payments, BCrypt password hashing
Reliability	Exponential backoff retry (1s → 5s → 15s, 3 attempts); InMemory Outbox pattern
Maintainability	Clean Architecture per service; Shared.Contracts library; consistent patterns
Observability	Structured logging via Serilog; correlation IDs across services; audit trails
Portability	Docker Compose for local development; each service containerized
Uptime Target	99.5% (development/self-learning project; production would target 99.9%+)

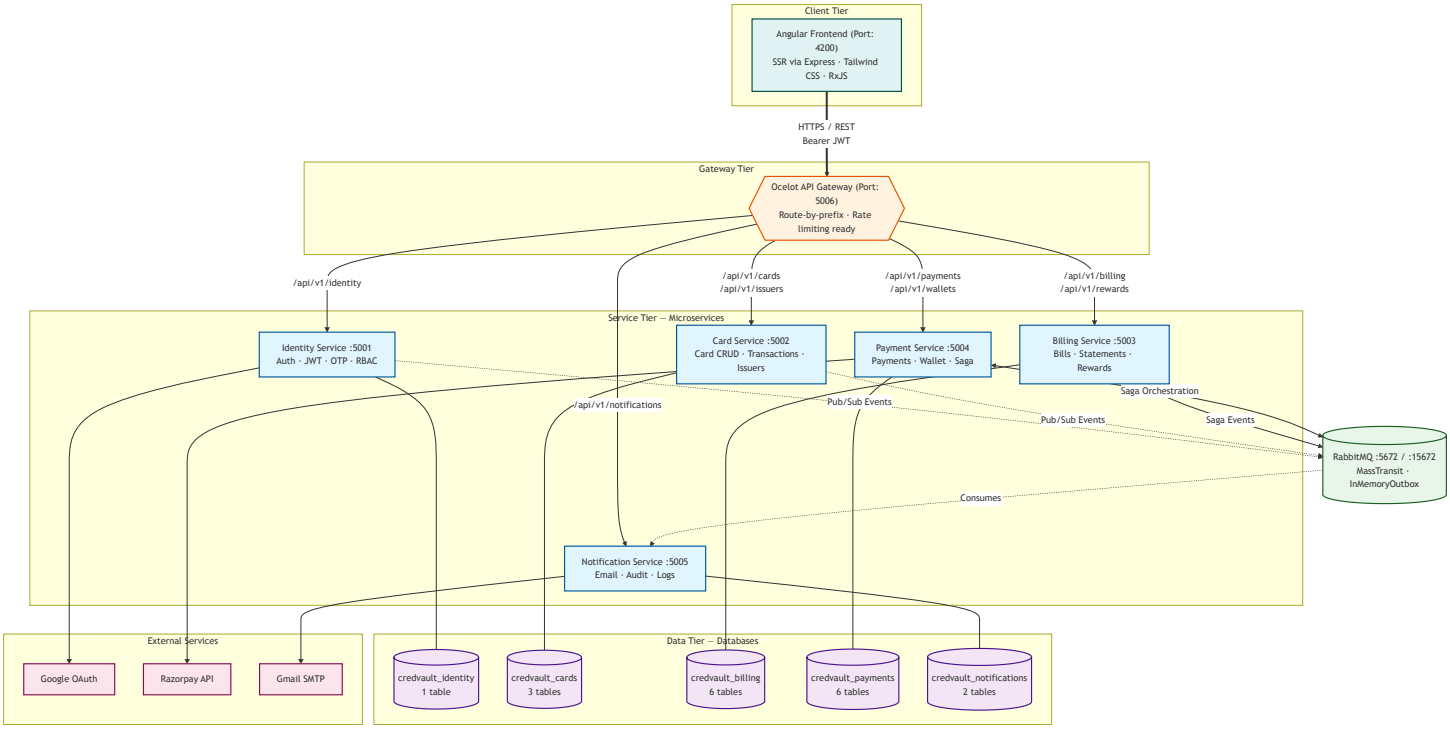
4. System Architecture

Architecture Style

Microservices with Event-Driven Communication + Saga Orchestration

The system is organized into five logical tiers with clear separation of concerns. Services communicate synchronously via the API Gateway (client-facing) and asynchronously via RabbitMQ (inter-service).

High-Level Architecture Diagram



Key Components

Component	Technology	Responsibility
Angular Frontend	Angular 21, TypeScript, Tailwind CSS	User interface, SSR via Express, JWT management in sessionStorage
Ocelot API Gateway	Ocelot (.NET)	Request routing by prefix, rate limiting, single entry point
Identity Service	.NET 8/10, EF Core, MediatR	User registration, authentication, JWT issuance, OTP lifecycle, RBAC
Card Service	.NET 8/10, EF Core, MediatR	Card CRUD, transaction recording, issuer management, AES encryption
Billing Service	.NET 8/10, EF Core, MediatR	Bill generation, statement creation, reward accounts, reward tiers
Payment Service	.NET 8/10, EF Core, MediatR, MassTransit	Payment flow, OTP 2FA, wallet operations, Saga orchestration, Razorpay

Component	Technology	Responsibility
Notification Service	.NET 8/10, EF Core, MassTransit	Email delivery (Gmail SMTP), audit logging, notification tracking
RabbitMQ	RabbitMQ 3.12, MassTransit	Async pub/sub messaging, saga event routing, retry policies
SQL Server 2022	SQL Server, EF Core Code-First	5 isolated databases, auto-migrations on startup

Service Responsibilities Matrix

Feature	Identity	Card	Billing	Payment	Notification
User Registration	**Primary**	—	—	—	Email
Authentication	**Primary**	Validates JWT	Validates JWT	Validates JWT	—
User Management	**Primary**	—	—	—	Audit
Card CRUD	—	**Primary**	—	Deduction (saga)	Email
Bill Generation	—	—	**Primary**	—	—
Statement Creation	—	—	**Primary**	—	—
Rewards System	—	—	**Primary**	Redeem (saga)	—
Payment Initiation	—	—	—	**Primary**	OTP Email
Wallet Management	—	—	—	**Primary**	—
Email Notifications	Publishes event	Publishes event	—	Publishes event	**Primary**

Feature	Identity	Card	Billing	Payment	Notification
Audit Logging	—	—	—	—	**Primary**

5. Data Flow

5.1 Communication Patterns

Pattern	Type	Protocol	Direction	Usage
Client → Gateway	Sync	HTTPS	External	All user-facing API calls
Gateway → Service	Sync	HTTP (internal)	Internal	Request routing by prefix
Service → Service (Pub/Sub)	Async	RabbitMQ/AMQP	Internal	Domain events (registration, card added)
Service → Service (Saga)	Async	RabbitMQ/AMQP	Internal	Distributed payment orchestration
Service → Database	Sync	TCP/SQL	Internal	Data persistence via EF Core
Service → External	Sync	HTTPS/SMTP	External	Google OAuth, Razorpay, Gmail

5.2 API Gateway Routing

Route Prefix	Destination Service	Port
`/api/v1/identity/`	Identity Service	5001
`/api/v1/cards/`	Card Service	5002
`/api/v1/issuers/`	Card Service	5002
`/api/v1/billing/`	Billing Service	5003
`/api/v1/rewards/`	Billing Service	5003
`/api/v1/payments/`	Payment Service	5004

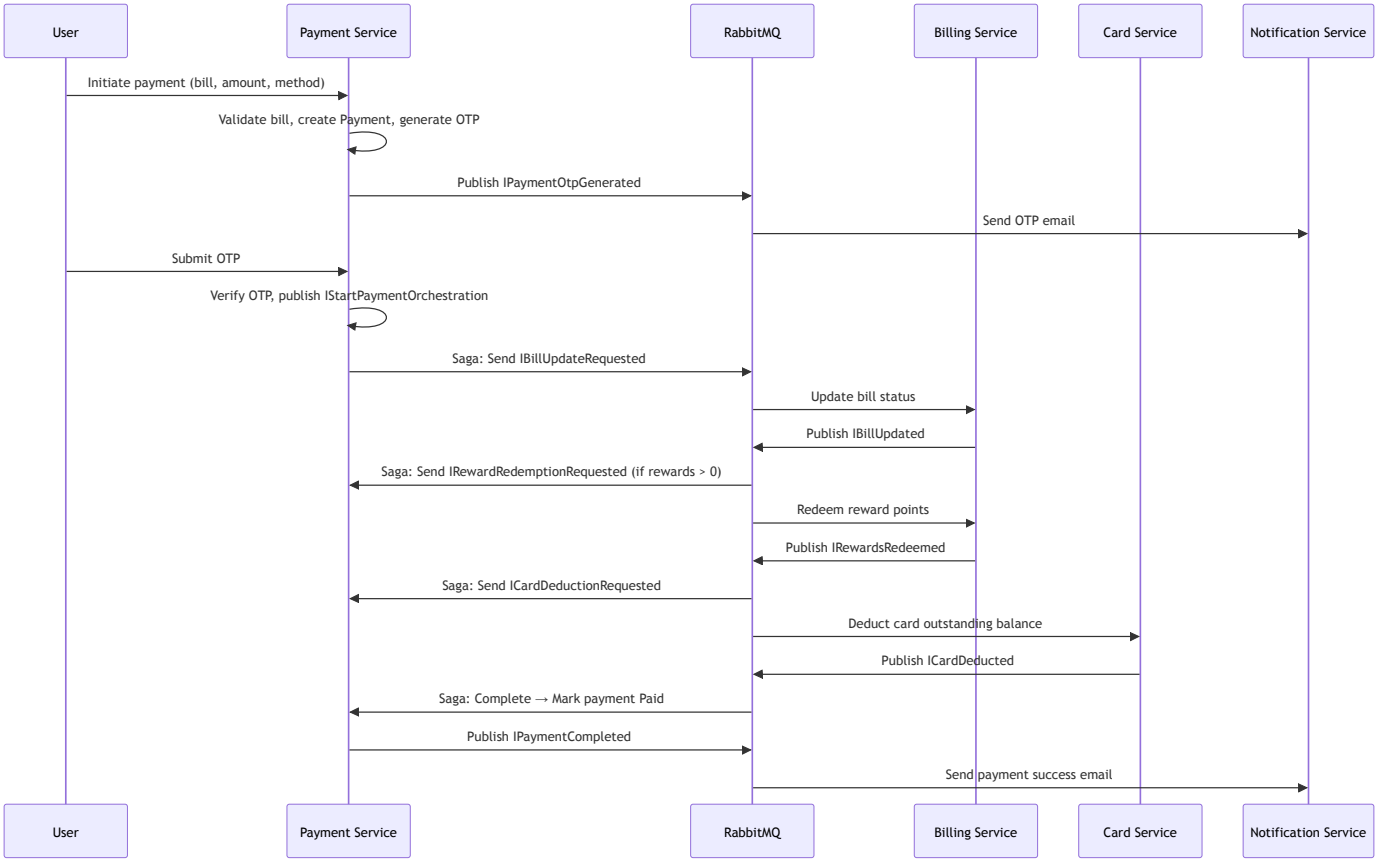
Route Prefix	Destination Service	Port
`/api/v1/wallets/`	Payment Service	5004
`/api/v1/notifications/`	Notification Service	5005

5.3 Event Routing

Event	Publisher	Consumer(s)	Purpose
`IUserRegistered`	Identity	Notification	Send welcome email
`IUserOtpGenerated`	Identity	Notification	Send OTP email
`ICardAdded`	Card	Notification	Send card confirmation email
`IPaymentOtpGenerated`	Payment	Notification	Send payment OTP email
`IStartPaymentOrchestration`	Payment	Payment (Saga)	Trigger payment saga
`IBillUpdateRequested`	Payment (Saga)	Billing	Update bill status
`IBillUpdated`	Billing	Payment (Saga)	Confirm bill update
`IRewardRedemptionRequested`	Payment (Saga)	Billing	Redeem reward points
`IRewardsRedeemed`	Billing	Payment (Saga)	Confirm reward redemption
`ICardDeductionRequested`	Payment (Saga)	Card	Deduct card balance
`ICardDeducted`	Card	Payment (Saga)	Confirm card deduction
`IPaymentCompleted`	Payment (Saga)	Notification	Send payment success email

Event	Publisher	Consumer(s)	Purpose
`IPaymentCompensated`	Payment (Saga)	Notification	Send payment failure email

5.4 Payment Saga Flow



6. Database Design (High-Level)

Database-per-Service Strategy

Each service owns its own isolated database. No cross-service foreign key constraints exist.

Database	Service	Tables	Key Entities
`credvault_identity`	Identity	1	`IdentityUser`
`credvault_cards`	Card	3	`CreditCard` , `CardTransaction` , `CardIssuer`

Database	Service	Tables	Key Entities
`credvault_billing`	Billing	6	`Bill` , `Statement` , `StatementTransaction` , `RewardAccount` , `RewardTier` , `RewardTransaction`
`credvault_payments`	Payment	6	`Payment` , `Transaction` , `UserWallet` , `WalletTransaction` , `PaymentOrchestrationSagaState` , `RazorpayWalletTopUp`
`credvault_notifications`	Notification	2	`AuditLog` , `NotificationLog`

Database Types

- **All databases**: SQL Server 2022 (relational)
- **ORM**: EF Core Code-First with auto-migrations on startup
- **Isolation**: Each service has its own DbContext; no shared tables
- **Consistency**: Eventual consistency via RabbitMQ events and Saga compensation

7. External Integrations

Integration	Service	Protocol	Purpose	Key Details
Google OAuth	Identity Service	HTTPS / OIDC	Passwordless SSO login	Validates Google IdToken with public keys; auto-creates user if new
Razorpay	Payment Service	HTTPS / REST	Wallet top-up payment processing	Creates orders, verifies HMAC-SHA256 webhook signatures, handles duplicates
Gmail SMTP	Notification Service	SMTP	Transactional email delivery	Used for OTP emails, welcome emails, card confirmations, payment notifications

8. Scalability Strategy

Horizontal Scaling

- All 5 microservices are **stateless** behind the API Gateway
- Each service can be scaled independently based on load
- Docker Compose supports multiple replicas per service

Load Balancing

- **Ocelot API Gateway** acts as the single entry point and load balancer
- Route-by-prefix configuration distributes requests to appropriate services
- Rate limiting capability ready for production

Caching Strategy

- **No distributed cache** (Redis) in current design — appropriate for self-learning scope
- CQRS pattern separates reads from writes, allowing future read-model caching
- JWT tokens cached client-side in sessionStorage (cleared on tab close)

Database Scaling

- Each service owns its database — can be moved to separate SQL Server instances
- EF Core query optimization via indexed foreign keys (e.g., `UserId` on CreditCards)

9. Reliability & Fault Tolerance

Retry Mechanisms

- **MassTransit consumers**: Exponential backoff — 1s → 5s → 15s (3 attempts)
- Applied to all event consumers across all services

Circuit Breakers

- Not explicitly implemented (self-learning scope)

- Service autonomy prevents cascade failures — if one service is down, others continue operating

Failover Strategy

- **Database per service**: Failure in one database doesn't affect others
- **Saga compensation**: If a distributed transaction step fails, all completed steps are reversed
- **Outbox pattern**: `UseInMemoryOutbox()` prevents message loss during service restarts
- **Idempotency**: Saga uses `CorrelationId` (GUID) as primary key; duplicate events are ignored

Saga Reliability

Mechanism	Configuration
Outbox	`UseInMemoryOutbox()` — prevents lost messages during restart
Retry	Exponential backoff: 1s → 5s → 15s (3 attempts)
Timeout	30 seconds per step; triggers compensation
Idempotency	CorrelationId (GUID) as saga PK; duplicates ignored
Persistence	`PaymentOrchestrationSagas` table tracks current state

10. Security Overview

Authentication

Aspect	Configuration
Token Type	JWT Bearer
Issuer	Identity Service
Audience	All services (shared config)
Expiry	60 minutes
Clock Skew	30 seconds tolerance
Token Payload	`sub` (UserId), `email`, `name`, `role`, `iss`, `aud`, `exp`

Aspect	Configuration
Client Storage	`sessionStorage` (cleared on tab close)

Authorization

Role	Access
Public	Registration, login, OTP verification, password reset
User	Own cards, bills, payments, wallet, profile
Admin	All user management, card management, bill generation, reward tiers, logs

Data Protection

Data	Protection
Card Numbers	AES encryption at rest; only last 4 digits stored plain
Passwords	BCrypt hashing (nullable for Google SSO users)
OTPs	6-digit codes with expiry (10 min for email, 5 min for payment)
API Transport	HTTPS for all external communication
Webhook Signatures	HMAC-SHA256 verification for Razorpay callbacks

11. Deployment Architecture

Cloud Provider

- **Current**: Docker Compose for local development
- **Future**: AWS/EKS or Azure AKS for production container orchestration

Docker Compose Services

```
services:
  identity-service:       # Port 5001 — User auth & management
  card-service:           # Port 5002 — Card CRUD & transactions
  billing-service:        # Port 5003 — Bills, statements, rewards
  payment-service:        # Port 5004 — Payments, wallet, saga
  notification-service:   # Port 5005 — Email, audit, logs
  gateway:                # Port 5006 — Ocelot API Gateway (exposed)
  rabbitmq:               # Port 5672, Management UI: 15672
  sqlserver:              # Port 1433 — All 5 databases
  frontend:               # Port 4200 — Angular SPA (dev)
```

Network Topology

Zone	Components	Accessibility
Public Zone	Gateway (5006), Frontend (4200)	Reachable from host/browser
Private Zone	All 5 microservices	Only reachable via Gateway or internal Docker DNS
Data Zone	SQL Server (1433), RabbitMQ (5672)	Only reachable by services on Docker bridge network

CI/CD Pipeline (Recommended)

- **Source Control**: GitHub
- **CI**: GitHub Actions — build, test, lint on every push
- **CD**: Docker image build → push to registry → deploy to Docker Compose or Kubernetes
- **Environments**: Dev (local Docker Compose), Staging, Production

Data Management

Aspect	Strategy
Migrations	EF Core `Database.Migrate()` on service startup
Seed Data	Card issuers (Visa, Mastercard, Rupay, Amex), default reward tiers

Aspect	Strategy
Persistence	Docker volumes for SQL Server data and RabbitMQ state
Internal DNS	Services communicate via Docker service names (e.g., `http://identity-api:80`)

12. Trade-offs & Decisions

Why Microservices Over Monolith?

- **Decision**: Microservices with 5 separate services
- **Rationale**: Demonstrates distributed system patterns (Saga, CQRS, event-driven); each service can be developed, tested, and deployed independently
- **Alternative**: Monolith would be simpler but wouldn't showcase enterprise patterns

Why RabbitMQ Over Kafka?

- **Decision**: RabbitMQ with MassTransit
- **Rationale**: Simpler setup for development; MassTransit provides excellent .NET integration with built-in saga support; sufficient for the message volume of this project
- **Alternative**: Kafka would be overkill for this scale and lacks native saga orchestration

Why SQL Server Over PostgreSQL/NoSQL?

- **Decision**: SQL Server 2022 for all databases
- **Rationale**: Strong EF Core support; ACID compliance needed for financial data; consistent technology across services simplifies development
- **Alternative**: PostgreSQL would work equally well; NoSQL not needed given relational data model

Why MassTransit InMemory Outbox Over Transactional Outbox?

- **Decision**: `UseInMemoryOutbox()`
- **Rationale**: Simpler implementation; adequate for development/self-learning; avoids additional database table per service

- **Alternative**: Transactional outbox with a separate OutboxEvents table would be more production-resilient

Why Saga Choreography Over Orchestration?

- **Decision**: MassTransit State Machine (orchestration-based)
- **Rationale**: Centralized state tracking in Payment Service makes debugging easier; clear compensation paths; state persisted for recovery
- **Alternative**: Pure choreography (event-based without central state) would be harder to trace and debug

Why No Redis/Distributed Cache?

- **Decision**: No caching layer
- **Rationale**: Read operations are not a bottleneck for a self-learning project; CQRS already separates reads from writes
- **Alternative**: Redis could be added later for read-heavy endpoints (e.g., user profile, reward balance)

End of High-Level Design Document